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Design, Build and Testing of a 3D Hologram-like RGB Fan Display

Mechatronic Project 478
Project Proposal

Teagan Reddy
26012243

Supervisor
Dr Melody Neaves

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sonke siya phambili
saam vorentoe

Department of Mechanical and Mechatronic Engineering
Stellenbosch University
Tel: +27 21 808 4204 | www.eng.sun.ac.za
Private Bag X1, Matieland, 7602, South Africa

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Executive Summary

Title of Project
Design, build and testing of a 3D hologram-like RGB fan display
Objectives
1. Formulate a set of specifications for an affordable 3D hologram-like RGB LED fan display unit 2. Generate and evaluate a range of concepts that align with the specifications 3. Select suitable and safe materials and components for the manufacturing of the final design 4. Develop and integrate a user-interface and the required software 5. Test the unit by using several visual display inputs of varying complexity (including static and dynamic scenes)
What is current practice and what are its limitations?
Modern 3D LED fan displays are primarily used in high-end advertising and digital signage. These units are expensive and rely on proprietary technology, which prevents their use for educational and recreational purposes. Commercial units often lack integrated safety features which is a concern for these displays when they are used in public spaces.
What is new in this project?
This project aims to reproduce the effect and functionality of a high-cost 3D hologram-like RGB LED fan display unit, while adhering to defined budgetary constraints.
If the project is successful, how will it make a difference?
This project will contribute to the use of 3D hologram-like displays by improving the quality and availability of hologram-like displays at the lower end of the market.
What are the risks to the project being a success? Why is it expected to be successful?
The risks of the project can be categorised into three parts: time constraints, budget constraints and the advanced hardware-to-software integration required. These risks are mitigated through strategies such as effective and detailed project planning to manage the time and budget constraints, and comprehensive market research to identify low-cost and efficient solutions to satisfy the component requirements.
What contributions have/will other students made/make?
A previous Stellenbosch student (Stefan Botes, 2025) has completed this project in the previous academic year.
Which aspects of the project will carry on after completion and why?
Future students will be able to investigate a multi-faceted design where multiple fans are integrated to allow for a larger, coordinated display unit. This will be possible as the functional base unit will already be developed.
What arrangements have been/will be made to expedite continuation?
The physical display unit will be manufactured as a modular and portable system to support any future hardware alterations. Furthermore, the developed software will be included in the project file.

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1. Introduction

1.1 Background

Advancements in visual display technology have significantly transformed the way information is presented in modern society, ranging from educational tools to commercial advertising platforms. Among emerging display technologies, hologram-like fan displays have attracted considerable attention due to their ability to create a mesmerising visual experience by allowing the three-dimensional (3D) rendering of objects in a physical space. Although these displays are often referred to as “holographic,” they do not rely on true holography. Instead, they exploit the persistence of vision (POV) phenomenon, a visual perception effect in which rapidly moving light sources create the illusion of continuous images to the human eye.

Persistence of vision occurs when the human eye retains an image for a short duration after the stimulus has disappeared (Paul et al., 2013). By rapidly rotating a strip or array of light-emitting diodes (LEDs) and precisely controlling their illumination timing, a two-dimensional (2D) image can be rendered in free space. When the rotational speed is sufficiently high, the discrete light pulses produced by the LEDs blend together, forming the perception of a stable and three-dimensional visual display. This technique enables the creation of static and dynamic images (Mishra et al., 2021).

Despite their visual appeal and growing popularity, commercially available hologram-like LED fan display units are often expensive and inaccessible for general use. Many existing systems are proprietary and rely on specialised hardware and software, limiting their flexibility and adaptability for custom applications. As a result, there is growing interest in the development of low-cost display units that can reproduce similar visual effects, while remaining accessible for educational demonstrations and recreational experimentation.

Furthermore, the unique floating and 3D visual effect produced by hologram-like LED fan displays creates a highly engaging viewing experience. This makes the technology particularly suitable for a variety of applications, including interactive classroom demonstrations and advertising displays (Fokides et al., 2023). The ability to present visually compelling graphics and animations in open space offers potential for enhancing audience engagement and information delivery (Ortega et al., 2020).

Additive manufacturing (AM) is a breakthrough technology that overcomes the geometrical limitations typically associated with traditional manufacturing routes, by taking information directly from computer-aided design (CAD) files (Wong and Hernandez, 2012). In this project, AM will be leveraged to produce a lightweight, balanced rotor and body design with integrated mounting channels for high-density RGB LED strips. This facilitates the design of complex aerodynamic geometries that ensure structural integrity and safety during high-speed rotation.

A previous Stellenbosch student (Stefan Botes, 2025) has completed this project in the previous academic year. However, this project will not be an extension or improvement of that implementation. Instead, this project will be developed from first principles using an original design process. Although, any previous contributions will be considered as a reference for research to analyse and identify any possible shortcomings.

Therefore, this project aims to design, build, and test a low-cost 3D hologram-like RGB LED fan display system that leverages the persistence of vision effect. The system will focus on achieving a balance between visual quality, affordability, portability, and energy efficiency, while also incorporating custom control software that enables the user to upload and display custom visual content. This work forms part of Mr Teagan Reddy's final year Mechatronics Engineering 478 Project. It will take place from 2 January 2026 until 6 November 2026. Experimental facilities within the Mechatronics and Electronics laboratory at Stellenbosch University will be used within a budget of R5000.

1.2 Objectives

The primary aim of this project is to design and construct a 3D hologram-like LED fan display unit, which is functional, safe and cost effective. The display unit will be designed and developed from first principles, and demonstrate applied mechatronic integration techniques. The following objectives have been defined to achieve this:

1. Formulate a set of specifications for an affordable 3D hologram-like RGB LED fan display unit
2. Generate and evaluate a range of concepts that align with the specifications
3. Select suitable and safe materials and components for the manufacturing of the final design
4. Develop and integrate a user-interface and the required software
5. Test the unit by using several visual display inputs of varying complexity (including static and dynamic scenes)

1.3 Motivation

Current commercial implementations of 3D hologram-like RGB LED fan display systems remain limited in accessibility due to their relatively high cost. While low-cost alternatives are available on the market, these options often compromise on important factors such as quality, operational safety, durability and overall performance.

This creates a gap between affordability and functionality in existing hologram-like display technologies. Developing a low-cost display unit would allow the application of hologram-like displays to be expanded into the educational and recreational sector.

Furthermore, the development of a low-cost display unit has potential benefits beyond educational use. An important advantage of LED fan displays lies in their potential energy efficiency. Conventional screens, such as LED panels or LCD displays, require millions of illuminated pixels. However, a rotating LED fan display utilises a single strip of LEDs attached to a rotating arm, which can generate similar images while consuming less power when compared to conventional screens. This characteristic makes such displays attractive for commercial applications, particularly in advertising environments where large, attention-grabbing displays are required but energy consumption must remain manageable (Chandrika et al., 2017).

Therefore this project aims to address the gap between cost and quality by designing and developing a functional, safe, and affordable 3D hologram-like LED fan display unit that can serve as an accessible platform for educational demonstrations and advertising applications.

2. Planned Activities

This section highlights the planned activities of the project. Furthermore, the time and cost estimates relating to each activity are provided in Appendix A.

2.1 Literature Review on Existing 3D Hologram LED Fan Displays and Alternative Implementations

A comprehensive literature review will be conducted into the current implementations of 3D hologram-like LED fans. Alternative implementations will also be researched to identify any possible innovation and cost reduction opportunities.

2.2 Define Stakeholder and Engineering Requirements

The system will be decomposed into a set of requirements that represent the target performance and success criterion. The defined requirements will guide the design process for the hardware-to-software integration, and allow for an objective project evaluation process.

2.3 Develop Hardware and Software Concepts

Concepts will be generated using the requirements from Activity 2.2 and information from Activity 2.1. To ensure compatibility, concepts will be generated for both the hardware design and software implementation .

2.4 Evaluate Concepts and Select Final Design

The concepts generated in Activity 2.3 will be objectively assessed with a decision matrix, based on the requirements from Activity 2.2. A final design concept will be chosen from the results of the decision matrix.

2.5 Research and Source the Required Components

Using the information from Activity 2.4, the components required for the final system will be researched and sourced. The hardware design and software implementation will also be considered when selecting components.

2.6 Develop Final Hardware Design

The hardware design will be directly integrated with the software development highlighted in Activity 2.7, to ensure compatibility. The hardware design will be completed using the information from Activity 2.4 and 2.5.

2.7 Develop Final Software Implementation

The software development will be directly integrated with the hardware implementation highlighted in Activity 2.6, to ensure compatibility. The software will be developed for the user interface and microcontroller that operates the 3D Hologram LED fan.

2.8 Acquire Required Components

The required components from Activity 2.5 and 2.6 will be acquired, and any required manufacturing will be completed.

2.9 Construct and Assemble the 3D Hologram LED Fan

The physical system will be constructed and assembled according to the final detailed design of Activity 2.6, using the components sourced and acquired in Activity 2.5 and 2.8 respectively.

2.10 Complete System Testing and Optimisation

The system will be tested against the requirements listed in Activity 2.2 to evaluate the success and shortcomings of the system. Furthermore, the system can be optimised and altered if any changes are required to meet the requirements.

2.11 Compile Report and Finalise Documentation

All relevant documentation and records of activities will be collected and compiled. A document repository will be developed during the course of the project. The final document will represent a comprehensive report of the project, including the design process, testing methods and end results.

3. Project Risk Assessment

The following section details the primary risks that could prevent the project from being successful and highlights the mitigation strategies that will be used to reduce the effect of these risks.

3.1 Time Constraints

Time constraints pose a prominent risk to the project being successful. Manufacturing, component sourcing, component acquisition and software development could lead to delays in the project timeline. Furthermore, if the requirements are found to be unsatisfied during the testing phase, this will lead to larger problems in the timeline towards the end of the project. However, a detailed and effective project plan will mitigate the risk of time constraints, especially if it contains considerations for the hardware-to-software integration and testing phase.

3.2 Budget Constraints

The second notable risk to the project being successful is budget constraints. The system requires advanced hardware-to-software integration, which could result in a need for proprietary parts and custom manufacturing. Furthermore, if these parts are unsuccessful in their application, the replacement costs would strain the budget even further. If the parts cannot be sourced from local manufacturers, and have to be purchased internationally, this would further increase the costs and lead time for components. It is possible to mitigate the risk of budget constraints with effective project planning and comprehensive market research, to ensure the components strike a balance between quality, requirement satisfaction and cost.

3.3 Advanced Hardware-to-Software Integration Requirements

The final risk to the project is the advanced hardware-to-software integration required for system functionality. The system requires a user interface that is able to control and provide inputs to the system. Additionally, the system requires advanced software to create static and dynamic scenes with a hologram-like effect. Furthermore, this system requires constant communication and speed control between the sensors and motors operating the system, while operating in a safe and quiet manner. All of these system parts are required to work together and be optimised for efficiency and performance. The mitigation strategy to manage the advanced hardware-to-software integration is to allocate sufficient time for development, construction, testing and optimisation in the project plan.

4. Conclusion

Current commercial 3D holographic displays are often prohibitively expensive and rely on closed-source software. This project facilitates the design, build and testing of a 3D hologram-like RGB fan display. Designing a custom unit from first principles, utilising additive manufacturing, offers a highly suitable route to overcome geometrical and cost limitations. Effective project planning will be utilised to manage the time and budget constraints, and ensure the project is delivered on time and to the specified requirements. The successful completion of this project will provide Stellenbosch University with a functional, open-access, and low-cost RGB LED fan display for educational and demonstrative purposes.

The completed project will also encourage further research into 3D hologram-like display units. The project is expected to take approximately 10 months to complete (concluding in November 2026) within a budget of R236 000. Running and material costs will be managed to remain within the allocated budget of R5 000.

5. References

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Appendix A: Planning Details

Appendix A details the estimated costs associated with the project activities as specified in Section 2. Table 1 summarises the estimated budget for planned activities and the total cost to complete the project. The allocated budget of R5 000 will be utilised for running costs and Mechanical and Mechatronic Workshop (MMW) material costs.

The engineering costs are based on a nominal rate of R500 per hour for a Junior Engineer. This rate corresponds to the assigned workload for the Mechanical and Mechatronic 478 Project module, which is 45 credits, where ten hours of work represent one credit. The estimated hours per activity are derived from the practical experience and theoretical knowledge gained throughout the first three years of an undergraduate engineering degree.

Running costs encompass the costs of goods and services that will be consumed throughout the project. The electrical running costs include microcontrollers, sensors, power supply, etc. The mechanical running costs include motors, supports, materials for manufacturing, etc.

Facility use typically accounts for the utilisation of previously acquired equipment or existing infrastructure, including associated depreciation and maintenance costs. However, as no existing infrastructure or equipment is required for this project, no expenses related to facility use have been included in the estimated budget.

Capital costs refer to expenditures incurred for the purchase of individual equipment exceeding R2 000, which will remain available for future use after the project has been completed. For the proposed project, items that could potentially fall under this category include high-cost microcontrollers, power transmitters, and motors.

Mechanical and Mechatronic Workshop (MMW) labour costs account for the time spent by workshop technicians in the manufacture of components required for the project. These costs are allocated at a rate of R300 per hour. MMW material costs refer to the materials consumed during the fabrication process, including items such as 3D printing filament used in additive manufacturing.

All expenses incurred outside the university, typically running, capital and material costs, include Value Added Tax (VAT).

Table 1 Budget for Planned Activities

Activity	Engineering Time		Running Costs	Facility Use	Capital Costs	MMW Labour		MMW Material	Total
	hr	R	R	R	R	hr	R	R	R
1. Literature Review	60	30 000	-	-	-	-	-	-	30 000
2. Define Requirements	30	15 000	-	-	-	-	-	-	15 000
3. Develop Concepts	45	22 500	-	-	-	-	-	-	22 500
4. Evaluate Concepts and Select Final Design	30	15 000	-	-	-	-	-	-	15 000
5. Research and Source Components	30	15 000	2 500	-	2 500	-	-	-	20 000
6. Develop Final Hardware Design	50	25 000	-	-	-	-	-	-	25 000
7. Develop Final Software Implementation	50	25 000	-	-	-	-	-	-	25 000
8. Acquire Components	20	10 000	2 000	-	-	-	-	-	12 000
9. Construction and Assembly	25	12 500	500	-	-	10	3 000	500	16 500
10. Testing and Optimisation	50	25 000	-	-	-	-	-	-	25 000
11. Compile Report and Finalise Documentation	60	30 000	-	-	-	-	-	-	30 000
TOTAL	450	225 000	5 000	0	2 500	10	3 000	500	236 000

Appendix B: Gantt Chart

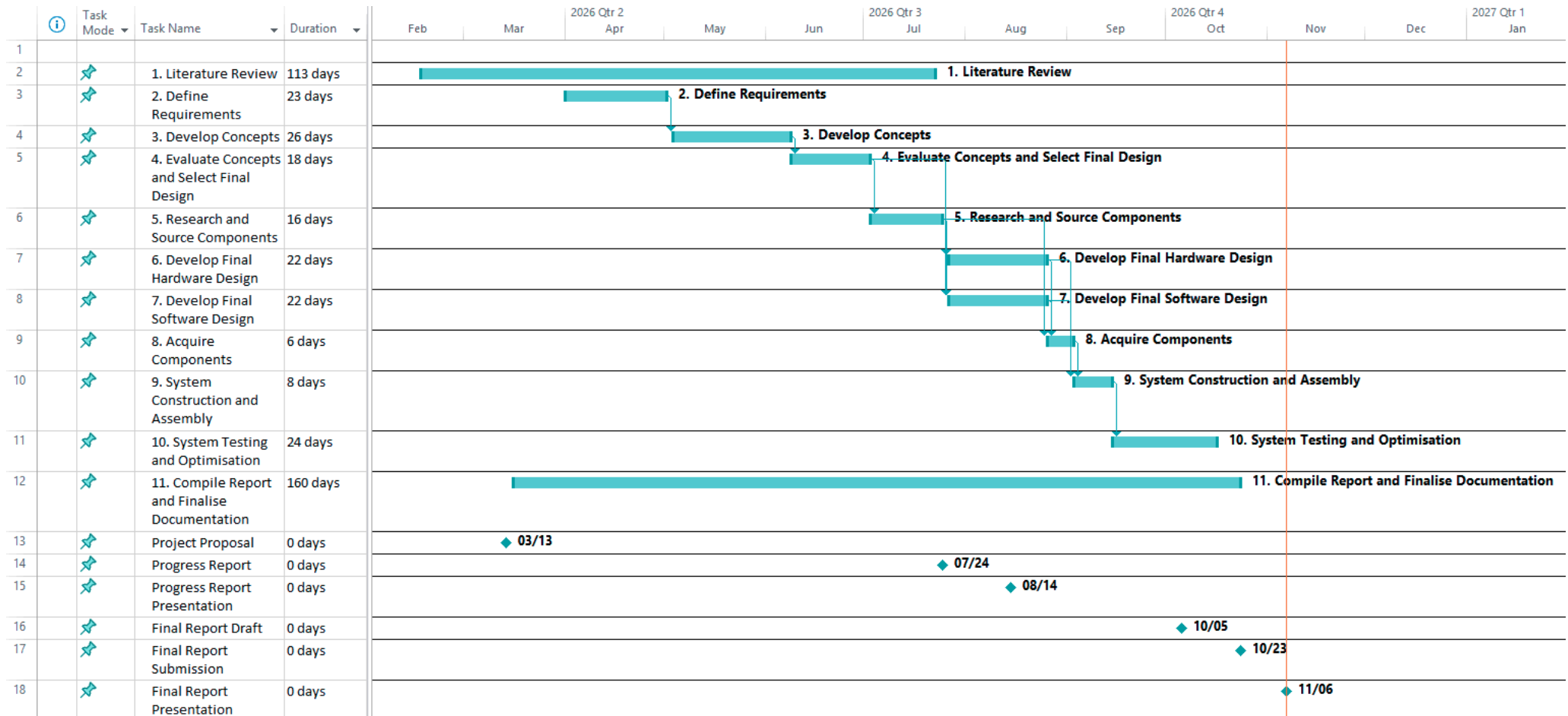


Figure 1 Gantt Chart of Planned Project Activities and Milestones

Appendix C: ECSA Graduate Attributes Self-Assessment

ECSA Graduate Attribute	Project Activities
GA 1: Problem Solving	The literature review in Activity 1 will identify limitations and possible improvements in the current implementations of 3D hologram LED fan displays. The information from Activity 1 and requirements from Activity 2 will require critical evaluation to generate and select concepts in Activity 3 and 4. Moreover, Activity 6 and 7 will require complex hardware-to-software integration; the system testing and optimisation, in Activity 10, demands critical thinking for troubleshooting and refinement.
GA 2: Application of Scientific and Engineering Knowledge	Basic engineering principles will be applied throughout the early stages of this project, including the requirements definition and concept generation process. This will require knowledge from Mechatronics 434 and Project Management 412. Advanced engineering principles will be required for the complex hardware-to-software integration. This will require application of knowledge drawn from Electronics 315, Computer Systems 245 and Design (E) 314. Furthermore, the design of a safe, quiet and balanced system will also require applied knowledge from Vibration and Noise 354 and Machine Design A 314.
GA 3: Engineering Design	The design concepts generated through the requirements analysis will explore a range of hardware and software solutions. In an engineering design context, these concepts will be critically evaluated and components will be thoroughly researched in Activities 4, 5, 6 and 7 respectively. The design choices must consider the interactions between hardware and software components, to create an operational system.
GA 5: Engineering Methods, Skills and Tools, including Information Technology	The generation of concept designs in Activity 3 and the final design development in Activity 6 will require the advanced use of an AutoCAD software, such as Autodesk Inventor Professional. The software development in Activity 7 will require the use of a computer package for computation, simulation and information handling. More specifically, the software development will involve using a range of programming tools and information technology to create a functional system. Lastly, the construction process in Activity 9 will need the skills and tools from Design (E) 314 and Practical Workshop Training 211.
GA 6: Professional and Technical Communication	Professional and technical communication will be practised, both orally and in writing, throughout each activity. Written information will be provided through the conventional methods of the discipline, such

	as engineering drawings for the concept design in Activity 3. Furthermore, all the information collected throughout the project will be compiled into a final document in Activity 11, and also explained through an oral presentation in a professional context.
GA 8: Individual, Team and Multidisciplinary Working	This project requires the demonstration of competence in independent work, as well as in a multidisciplinary context. Although the project is completed individually, it incorporates multidisciplinary work through the integration of multiple engineering disciplines, including project management, mechanical and electrical engineering. The development of a functional system requires the application and coordination of knowledge across these disciplinary boundaries, thereby reflecting the collaborative and integrative nature of multidisciplinary engineering practice.
GA 9: Independent Learning Ability	Independent learning within a complex defined context will be demonstrated throughout this project. The design process in Activity 3 requires knowledge and technical skills beyond prior coursework, based on the literature review in Activity 1, where academic resources and technical documentation will be reviewed and critically evaluated. Furthermore, the broader social and ethical implications of applying engineering knowledge will be considered in this project, including safe and responsible design practices, and the impact of system performance on users and stakeholders.

Table 2 ECSA Graduate Attributes Self-Assessment

Appendix D: Responsible Resource Use and End-of-Life Strategy

This appendix outlines the strategy for the responsible resource management and end-of-life treatment throughout the project lifespan. Resource management includes considerations for responsible sourcing and environmental impact during design, development and testing. The end-of-life strategy is guided by sustainability, where component reusability and recyclability are prioritised. Components are intended to be repurposed for future academic or development projects where possible. A summary of the resource management and end-of-life strategies is provided in Table 3.

Resource Category	Responsible Acquisition and Use	End-of-Life Strategy
Electrical Components	Components will be sourced through university-accredited suppliers to ensure quality assurance. Preference will be given to local suppliers in order to reduce transportation distances and the associated carbon emissions. Moreover, components that are available from the same supplier will be consolidated into a single order to minimise packaging waste and multiple delivery cycles. International suppliers will only be considered when specialised or unavailable components are required. This resource acquisition strategy reduces the environmental impact associated with long distance transportation, while promoting responsible and efficient resource utilisation.	Electrical components, including motors, sensors, and microcontrollers, will be selected with the intention of reuse in future academic or development projects. Components will be carefully handled, tested, and stored to preserve functionality and extend their operational lifespan. Components that become damaged beyond repair during the project will be directed to appropriate electronic waste recycling facilities, in accordance with environmental regulations. This approach minimises the generation of electronic waste (E-Waste) and supports responsible end-of-life management of electronic equipment.
Mechanical Components	Mechanical components will be manufactured primarily within the university workshop. Recyclable and sustainably sourced materials will be selected to reduce environmental impact and	Used mechanical components will be assessed for reuse in suitable future projects. Any components that remain functional, will be stored for subsequent academic or development projects.

	support responsible resource management. Specialised components that cannot be manufactured in the workshop using conventional methods, will be produced through additive manufacturing. This method reduces material waste and allows efficient production of complex geometries.	If components cannot be reused, they will be handled in the following way: Recyclable materials, including additive manufacturing polymers, will be directed to the approved recycling facilities. Materials that cannot be recycled will be disposed of in accordance with the relevant waste management and environmental regulations.
Software	Software development will prioritise efficient use of computational resources, including the optimised use of processing power and memory storage. Code will be structured to minimise computational overhead, to reduce energy consumption and improve system performance. All developed software will be included in the final project repository to facilitate reuse and future development. These methods for software development promote longevity, transparency and maintainability.	When the software is no longer required for operational use, it will be archived using repository management tools. This ensures that the software development information is preserved for future reference and development. Redundant files will be securely removed to optimise storage usage and maintain efficient system resource management.

Table 3 Resource Use Plan and End-of-Life Strategy

